

CSE420: Compiler Design

Self Study Assignment 02

Submit with a cover page containing your Name, ID,
Section (MUST)
Deadline: 4th Jan

- Given the SDT, evaluate the following string with either annotated parse tree or dependency graph

String: **record { int [2][6] a; float [7][2] b; } x; int y;**

SDT:

$P \rightarrow \{ \text{offset} = 0; \} D$	
$D \rightarrow T \text{ id } ; \{ \text{top.put}(\text{id.lexeme}, T.type, \text{offset}); D_1$ $\quad \quad \quad \text{offset} = \text{offset} + T.width; \}$	
$D \rightarrow \epsilon$	
$T \rightarrow \text{record '}' \{ \text{Env.push}(\text{top}); \text{top} = \text{new Env}(); \quad D \text{ '}'$ $\quad \quad \quad \text{Stack.push}(\text{offset}); \text{offset} = 0; \}$	$\{ T.type = \text{record}(\text{top}); T.width = \text{offset};$ $\quad \quad \quad \text{top} = \text{Env.pop}(); \text{offset} = \text{Stack.pop}(); \}$
$T \rightarrow B \{ t = B.type; w = B.width; \} C$	$\{ T.type = C.type; T.width = C.width; \}$
$B \rightarrow \text{int}$	$\{ B.type = \text{integer}; B.width = 4; \}$
$B \rightarrow \text{float}$	$\{ B.type = \text{float}; B.width = 8; \}$
$C \rightarrow \epsilon$	$\{ C.type = t; C.width = w; \}$
$C \rightarrow [\text{num}] C_1$	$\{ C.type = \text{array}(\text{num.value}, C_1.type);$ $\quad \quad \quad C.width = \text{num.value} \times C_1.width; \}$